EXPERIMENT - 15

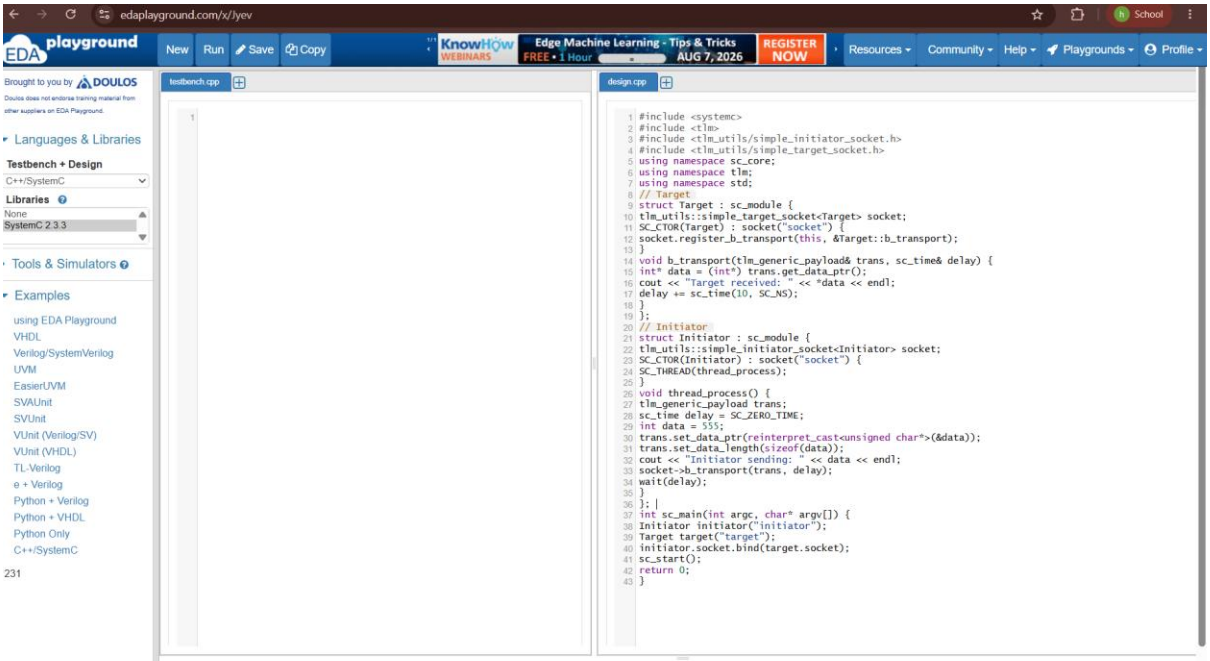
TRANSACTION LEVEL MODELING (TLM) USING SYSTEMC

CODE:

```
#include <systemc>
#include <tlm>
#include <tlm_utils/simple_initiator_socket.h>
#include <tlm_utils/simple_target_socket.h>
using namespace sc_core;
using namespace tlm;
using namespace std;
// Target
struct Target : sc_module {
tlm_utils::simple_target_socket<Target> socket;
SC_CTOR(Target) : socket("socket") {
socket.register_b_transport(this, &Target::b_transport);
}
void b_transport(tlm_generic_payload& trans, sc_time& delay) {
int* data = (int*) trans.get_data_ptr();
cout << "Target received: " << *data << endl;
delay += sc_time(10, SC_NS);
}
};
// Initiator
struct Initiator : sc_module {
tlm_utils::simple_initiator_socket<Initiator> socket;
```

```
SC_CTOR(Initiator) : socket("socket") {
SC_THREAD(thread_process);
}
void thread_process() {
tlm_generic_payload trans;
sc_time delay = SC_ZERO_TIME;
int data = 555;
trans.set_data_ptr(reinterpret_cast<unsigned char*>(&data));
trans.set_data_length(sizeof(data));
cout << "Initiator sending: " << data << endl;
socket->b_transport(trans, delay);
wait(delay);
}
};
int sc_main(int argc, char* argv[]) {
Initiator initiator("initiator");
Target target("target");
initiator.socket.bind(target.socket);
sc_start();
return 0;
}
```

SIMULATION:



OUTPUT: